

School of Computer Science and Artificial Intelligence

Lab Assignment # 11.1

Program	: B. Tech (CSE)
Specialization	: -
Course Title	: AI Assisted Coding
Course Code	: 23CS002PC304
Semester	: II
Academic Session	: 2025-2026
Name of Student	: K.Dhanush
Enrollment No.	: 2403A51L32
Batch No.	: 52
Date	: 17/02/26

Submission Starts here**Screenshots:****Task Description #1 – Stack Implementation**

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample code:

```
class Stack:  
    Pass
```

Prompt:

generate a Stack class with push, pop, peek, and is_empty
Methods and code.

Code:

```

1 class Stack:
2     def __init__(self):
3         """Initialize an empty stack."""
4         self.items = []
5
6     def push(self, item):
7         """Add an item to the top of the stack.
8
9         Args:
10            item: The item to be added to the stack.
11        """
12        self.items.append(item)
13
14    def pop(self):
15        """Remove and return the top item from the stack.
16
17        Returns:
18            The top item from the stack.
19
20        Raises:
21            IndexError: If the stack is empty.
22        """
23        if self.is_empty():
24            raise IndexError("Cannot pop from an empty stack")
25        return self.items.pop()
26
27    def peek(self):
28        """Return the top item from the stack without removing it.
29
30        Returns:
31            The top item from the stack.
32
33        Raises:
34            IndexError: If the stack is empty.
35        """
36        if self.is_empty():
37            raise IndexError("Cannot peek at an empty stack")
38        return self.items[-1]
39
40    def is_empty(self):
41        """Check if the stack is empty.
42
43        Returns:
44            True if the stack is empty, False otherwise.
45        """
46        return len(self.items) == 0
47
48    def size(self):
49        """Return the number of items in the stack.
50
51        Returns:
52            The number of items in the stack.
53        """
54        return len(self.items)
55
56    def __str__(self):
57        """Return a string representation of the stack.
58
59        Returns:
60            A string representation of the stack.
61        """
62        return f"Stack({self.items})"

```

```

63
64
65 # Example usage and testing
66 if __name__ == "__main__":
67     # Create a new stack
68     stack = Stack()
69
70     # Test is_empty on empty stack
71     print(f"Is stack empty? {stack.is_empty()}")
72     print(f"Stack size: {stack.size()}")
73     print()
74
75     # Push items onto the stack
76     print("Pushing items: 10, 20, 30")
77     stack.push(10)
78     stack.push(20)
79     stack.push(30)
80     print(f"Stack: {stack}")
81     print(f"Is stack empty? {stack.is_empty()}")
82     print(f"Stack size: {stack.size()}")
83     print()
84
85     # Peek at the top item
86     print(f"Peek at top: {stack.peak()}")
87     print(f"Stack after peek: {stack}")
88     print()
89
90     # Pop items from the stack
91     print("Popping items:")
92     print(f"Popped: {stack.pop()}")
93     print(f"Stack: {stack}")
94     print(f"Popped: {stack.pop()}")
95     print(f"Stack: {stack}")
96     print(f"Popped: {stack.pop()}")
97     print(f"Stack: {stack}")
98     print()
99
100     # Test is_empty after popping all items
101     print(f"Is stack empty? {stack.is_empty()}")
102     print(f"Stack size: {stack.size()}")

```

Output:

```

Is stack empty? True
Stack size: 0

Pushing items: 10, 20, 30
Stack: Stack([10, 20, 30])
Is stack empty? False
Stack size: 3

Peek at top: 30
Stack after peek: Stack([10, 20, 30])

Popping items:
Popped: 30
Stack: Stack([10, 20])
Popped: 20
Stack: Stack([10])
Popped: 10
Stack: Stack([])

Is stack empty? True
Stack size: 0
PS C:\2403A51L03\3-2\AI_A_C\Cursor AI>

```

Task Description #2 – Queue Implementation

Task: Use AI to implement a Queue using Python lists.

Sample Input Code:

```
class Queue:
```

Pass

Prompt:

complete the queue class with enqueue, dequeue, peek, and size methods and code

Code:

```

1  class Queue:
2      def __init__(self):
3          """Initialize an empty queue."""
4          self.items = []
5
6      def enqueue(self, item):
7          """Add an item to the rear of the queue.
8
9          Args:
10             item: The item to add to the queue
11          """
12          self.items.append(item)
13
14      def dequeue(self):
15          """Remove and return the front item from the queue.
16
17          Returns:
18             The front item of the queue
19
20          Raises:
21             IndexError: If the queue is empty
22          """
23          if self.is_empty():
24              raise IndexError("Cannot dequeue from an empty queue")
25          return self.items.pop(0)
26
27      def peek(self):
28          """Return the front item of the queue without removing it.
29
30          Returns:
31             The front item of the queue
32
33          Raises:
34             IndexError: If the queue is empty
35          """
36          if self.is_empty():
37              raise IndexError("Cannot peek at an empty queue")
38          return self.items[0]
39
40      def size(self):
41          """Return the number of items in the queue.
42
43          Returns:
44             The number of items in the queue
45          """
46          return len(self.items)
47
48      def is_empty(self):
49          """Check if the queue is empty.
50
51          Returns:
52             True if the queue is empty, False otherwise
53          """
54          return len(self.items) == 0
55
56
57 # Example usage and testing
58 if __name__ == "__main__":
59     # Create a new queue
60     q = Queue()
61
62     # Test enqueue
63     print("Enqueueing items: 1, 2, 3, 4, 5")
64     q.enqueue(1)
65     q.enqueue(2)
66     q.enqueue(3)
67     q.enqueue(4)
68     q.enqueue(5)
69
70     # Test size
71     print(f"Queue size: {q.size()}")
72
73     # Test peek
74     print(f"Peek at front: {q.peek()}")
75
76     # Test dequeue
77     print("\nDequeueing items:")
78     while not q.is_empty():
79         print(f"  Dequeued: {q.dequeue()}, Remaining size: {q.size()}")
80
81     # Test empty queue
82     print(f"Queue is empty: {q.is_empty()}")
83
84     # Test error handling
85     try:
86         q.dequeue()
87     except IndexError as e:
88         print(f"Error caught: {e}")
89
90     try:
91         q.peek()
92     except IndexError as e:
93         print(f"Error caught: {e}")

```

Output:

```

Enqueueing items: 1, 2, 3, 4, 5
Queue size: 5
Peek at front: 1

Dequeuing items:
Dequeued: 1, Remaining size: 4
Dequeued: 2, Remaining size: 3
Dequeued: 3, Remaining size: 2
Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Enqueueing items: 1, 2, 3, 4, 5
Queue size: 5
Peek at front: 1
0
Dequeuing items:
Dequeued: 1, Remaining size: 4
Dequeued: 2, Remaining size: 3
Dequeued: 3, Remaining size: 2
Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Peek at front: 1

Dequeuing items:
Dequeued: 1, Remaining size: 4
Dequeued: 2, Remaining size: 3
Dequeued: 3, Remaining size: 2
Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Dequeuing items:
Dequeued: 1, Remaining size: 4
Dequeued: 2, Remaining size: 3
Dequeued: 3, Remaining size: 2
Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Dequeuing items:
Dequeued: 1, Remaining size: 4
Dequeued: 2, Remaining size: 3
Dequeued: 3, Remaining size: 2
Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Dequeued: 2, Remaining size: 3
Dequeued: 3, Remaining size: 2
Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Dequeued: 4, Remaining size: 1
Dequeued: 5, Remaining size: 0

Dequeued: 5, Remaining size: 0

Queue is empty: True
Error caught: Cannot dequeue from an empty queue
Error caught: Cannot peek at an empty queue

```

Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and display methods.

Sample Input Code:

class Node:

Pass

Prompt:

generate a Singly Linked List with insert and display methods with code

Code:

```

1 class Node:
2     """Node class to represent a single node in the linked list"""
3     def __init__(self, data):
4         self.data = data # Data stored in the node
5         self.next = None # Reference to the next node
6
7
8 class SinglyLinkedList:
9     """Singly Linked List implementation with insert and display methods"""
10
11     def __init__(self):
12         self.head = None # Head pointer pointing to the first node
13
14     def insert(self, data):
15         """
16         Insert a new node at the end of the linked list
17
18         Args:
19             data: The data to be inserted into the linked list
20         """
21         new_node = Node(data)
22
23         # If the List is empty, make the new node the head
24         if self.head is None:
25             self.head = new_node
26         else:
27             # Traverse to the end of the List
28             current = self.head
29             while current.next is not None:
30                 current = current.next
31             # Insert the new node at the end
32             current.next = new_node
33
34     def insert_at_beginning(self, data):
35         """
36         Insert a new node at the beginning of the linked list
37
38         Args:
39             data: The data to be inserted into the linked list
40         """
41         new_node = Node(data)
42         new_node.next = self.head
43         self.head = new_node
44
45     def display(self):
46         """
47         Display all elements in the linked list
48         """
49         if self.head is None:
50             print("Linked List is empty")
51             return
52
53         current = self.head
54         elements = []
55         while current is not None:
56             elements.append(str(current.data))
57             current = current.next
58
59         # Display in format: data1 -> data2 -> data3 -> None
60         print(" -> ".join(elements) + " -> None")
61
62
63 # Example usage
64 if __name__ == "__main__":
65     # Create a new linked list
66     ll = SinglyLinkedList()
67
68     # Insert some elements
69     print("Inserting elements into the linked list...")
70     ll.insert(10)
71     ll.insert(20)
72     ll.insert(30)
73     ll.insert(40)
74
75     # Display the linked list
76     print("\nLinked List contents:")
77     ll.display()
78
79     # Insert at beginning
80     print("\nInserting 5 at the beginning...")
81     ll.insert_at_beginning(5)
82     ll.display()
83
84     # Create an empty list
85     print("\nCreating an empty linked list:")
86     empty_ll = SinglyLinkedList()
87     empty_ll.display()

```

Output:

```
Inserting elements into the linked list...
Inserting elements into the linked list...
```

```
Linked List contents:
```

```
Linked List contents:
```

```
Linked List contents:
```

```
10 -> 20 -> 30 -> 40 -> None
```

```
10 -> 20 -> 30 -> 40 -> None
```

```
Inserting 5 at the beginning...
```

```
Inserting 5 at the beginning...
```

```
5 -> 10 -> 20 -> 30 -> 40 -> None
```

```
5 -> 10 -> 20 -> 30 -> 40 -> None
```

```
Creating an empty linked list:
```

```
Linked List is empty
```

```
Linked List is empty
```

Task Description #4 – Binary Search Tree (BST)

Task: Use AI to create a BST with insert and in-order traversal methods.

Sample Input Code:

```
class BST:
    pass
```

Prompt:

create a BST with insert and in-order traversal methods and code

Code:

```

1 class Node:
2     def __init__(self, key: int):
3         self.key = key
4         self.left: "Node | None" = None
5         self.right: "Node | None" = None
6
7
8 class BST:
9     def __init__(self):
10        self.root: Node | None = None
11
12    def insert(self, key: int) -> None:
13        """Insert key into the BST (duplicates go to the right)."""
14        if self.root is None:
15            self.root = Node(key)
16            return
17
18        cur = self.root
19        while True:
20            if key < cur.key:
21                if cur.left is None:
22                    cur.left = Node(key)
23                    return
24                cur = cur.left
25            else:
26                if cur.right is None:
27                    cur.right = Node(key)
28                    return
29                cur = cur.right
30
31    def inorder(self) -> list[int]:
32        """Return keys in in-order (sorted) order."""
33        result: list[int] = []
34
35        def dfs(n: Node | None) -> None:
36            if n is None:
37                return
38            dfs(n.left)
39            result.append(n.key)
40            dfs(n.right)
41
42        dfs(self.root)
43        return result
44
45
46 if __name__ == "__main__":
47     bst = BST()
48     for x in [7, 3, 9, 1, 5, 8, 10]:
49         bst.insert(x)
50     print("In-order:", bst.inorder())

```

Output:

```
In-order: [1, 3, 5, 7, 8, 9, 10]
```

Task Description #5 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Sample Input Code:

```
class HashTable:
    pass
```

Prompt:

implement a hash table with basic insert, search, and delete methods with code

Code:


```

1 class HashTable:
2     """
3     Hash table using separate chaining (list of buckets).
4
5     Methods:
6     - insert(key, value): add/update a key
7     - search(key): return value or None if not found
8     - delete(key): remove key, return True if removed else False
9     """
10
11     def __init__(self, capacity: int = 8) -> None:
12         if capacity < 1:
13             raise ValueError("capacity must be >= 1")
14         self._capacity = capacity
15         self._buckets = [[] for _ in range(self._capacity)] # List[list[tuple[key, value]]]
16         self._size = 0
17
18     def _index(self, key) -> int:
19         return hash(key) % self._capacity
20
21     def _rehash(self, new_capacity: int) -> None:
22         old_items = []
23         for bucket in self._buckets:
24             old_items.extend(bucket)
25
26         self._capacity = new_capacity
27         self._buckets = [[] for _ in range(self._capacity)]
28         self._size = 0
29
30         for k, v in old_items:
31             self.insert(k, v)
32
33     def insert(self, key, value) -> None:
34         # Resize when Load factor gets too high (simple rule-of-thumb)
35         if (self._size + 1) / self._capacity > 0.75:
36             self._rehash(self._capacity * 2)
37
38         idx = self._index(key)
39         bucket = self._buckets[idx]
40
41         for i, (k, _) in enumerate[Any](bucket):
42             if k == key:
43                 bucket[i] = (key, value) # update existing
44                 return
45
46         bucket.append((key, value))
47         self._size += 1
48
49     def search(self, key):
50         idx = self._index(key)
51         bucket = self._buckets[idx]
52         for k, v in bucket:
53             if k == key:
54                 return v
55         return None
56
57     def delete(self, key) -> bool:
58         idx = self._index(key)
59         bucket = self._buckets[idx]
60
61         for i, (k, _) in enumerate[Any](bucket):
62             if k == key:
63                 bucket.pop(i)
64                 self._size -= 1
65                 return True
66
67         return False
68
69     def __len__(self) -> int:
70         return self._size
71
72     def __contains__(self, key) -> bool:
73         return self.search(key) is not None
74
75     def __repr__(self) -> str:
76         return f"HashTable(size={self._size}, capacity={self._capacity})"
77
78
79 if __name__ == "__main__":
80     ht = HashTable()
81     ht.insert("name", "Alice")
82     ht.insert("age", 20)
83     ht.insert("age", 21) # update
84
85     print(ht) # HashTable(...)
86     print(ht.search("name")) # Alice
87     print(ht.search("age")) # 21
88     print(ht.search("x")) # None
89
90     print(ht.delete("age")) # True
91     print(ht.delete("age")) # False
92     print(len(ht)) # 1

```

Output:

```
HashTable(size=2, capacity=8)
Alice
21
None
HashTable(size=2, capacity=8)
Alice
21
None
21
None
True
False
1
True
False
1
False
1
```

Task Description #6 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

```
class Graph:
    pass
```

Prompt:

implement a graph using an adjacency list with code

Code:

```

1 class Graph:
2     """
3     Graph implemented using an adjacency list.
4
5     - By default the graph is undirected.
6     - Set directed=True for a directed graph.
7     """
8
9     def __init__(self, directed: bool = False):
10         self.directed = directed
11         # adjacency list: vertex -> set of neighbor vertices
12         self.adj: dict[object, set[object]] = {}
13
14     def add_vertex(self, v: object) -> None:
15         """Add a vertex if it doesn't already exist."""
16         if v not in self.adj:
17             self.adj[v] = set[object]()
18
19     def add_edge(self, u: object, v: object) -> None:
20         """Add an edge u -> v (and v -> u if undirected)."""
21         self.add_vertex(u)
22         self.add_vertex(v)
23         self.adj[u].add(v)
24         if not self.directed:
25             self.adj[v].add(u)
26
27     def remove_edge(self, u: object, v: object) -> None:
28         """Remove an edge u -> v (and v -> u if undirected), if present."""
29         if u in self.adj:
30             self.adj[u].discard(v)
31         if not self.directed and v in self.adj:
32             self.adj[v].discard(u)
33
34     def remove_vertex(self, v: object) -> None:
35         """Remove a vertex and all edges incident to it."""
36         if v not in self.adj:
37             return
38
39         # Remove edges from neighbors to v
40         for n in list(self.adj[v]):
41             self.remove_edge(v, n)
42
43         # In directed graphs, also remove incoming edges to v
44         if self.directed:
45             for u in self.adj:
46                 self.adj[u].discard(v)
47
48         del self.adj[v]
49
50     def neighbors(self, v: object) -> list[object]:
51         """Return neighbors of v as a sorted list when possible."""
52         if v not in self.adj:
53             return []
54         try:
55             return sorted(self.adj[v])
56         except TypeError:
57             return list(self.adj[v])
58
59     def bfs(self, start: object) -> list[object]:
60         """Breadth-first traversal order starting from start."""
61         if start not in self.adj:
62             return []
63
64         visited = {start}
65         queue = [start]
66         order: list[object] = []
67
68         while queue:
69             v = queue.pop(0)
70             order.append(v)
71             for n in self.neighbors(v):
72                 if n not in visited:
73                     visited.add(n)
74                     queue.append(n)
75
76         return order
77
78     def dfs(self, start: object) -> list[object]:
79         """Depth-first traversal order starting from start."""
80         if start not in self.adj:
81             return []
82
83         visited: set[object] = set[object]()
84         order: list[object] = []
85
86         def _visit(v: object) -> None:
87             visited.add(v)
88             order.append(v)
89             for n in self.neighbors(v):
90                 if n not in visited:
91                     _visit(n)
92
93         _visit(start)
94         return order
95
96     def __str__(self) -> str:
97         lines = []
98         for v in self.adj:
99             lines.append(f"{v} -> {self.neighbors(v)}")
100         return "\n".join(lines)
101
102
103 if __name__ == "__main__":
104     g = Graph(directed=False) # change to True for a directed graph
105     g.add_edge("A", "B")
106     g.add_edge("A", "C")
107     g.add_edge("B", "D")
108     g.add_edge("C", "D")
109     g.add_edge("D", "E")
110
111     print("Adjacency list:")
112     print(g)
113     print("BFS from A:", g.bfs("A"))
114     print("DFS from A:", g.dfs("A"))

```

Output:

```
Adjacency list:
A -> ['B', 'C']
B -> ['A', 'D']
Adjacency list:
A -> ['B', 'C']
B -> ['A', 'D']
A -> ['B', 'C']
B -> ['A', 'D']
B -> ['A', 'D']
C -> ['A', 'D']
D -> ['B', 'C', 'E']
E -> ['D']

BFS from A: ['A', 'B', 'C', 'D', 'E']
DFS from A: ['A', 'B', 'D', 'C', 'E']
D -> ['B', 'C', 'E']
E -> ['D']

BFS from A: ['A', 'B', 'C', 'D', 'E']
DFS from A: ['A', 'B', 'D', 'C', 'E']
E -> ['D']

BFS from A: ['A', 'B', 'C', 'D', 'E']
DFS from A: ['A', 'B', 'D', 'C', 'E']
BFS from A: ['A', 'B', 'C', 'D', 'E']
DFS from A: ['A', 'B', 'D', 'C', 'E']
DFS from A: ['A', 'B', 'D', 'C', 'E']
```

Task Description #7 – Priority Queue

Task: Use AI to implement a priority queue using Python's heapq module.

Sample Input Code:

```
class PriorityQueue:
    pass
```

Prompt:

implement a priority queue using Python's heapq module with code

Code:

```

1 import heapq
2 from itertools import count
3
4 class PriorityQueue:
5     """
6     Min-priority queue by default (smaller priority value = served first).
7     For max-priority behavior, push with -priority.
8     """
9     def __init__(self):
10         self._heap = []
11         self._seq = count(int)() # tie-breaker for equal priorities (FIFO)
12
13     def push(self, item, priority: int):
14         heapq.heappush(self._heap, (priority, next(self._seq), item))
15
16     def pop(self):
17         if not self._heap:
18             raise IndexError("pop from empty PriorityQueue")
19         priority, _, item = heapq.heappop(self._heap)
20         return item, priority
21
22     def peek(self):
23         if not self._heap:
24             raise IndexError("peek from empty PriorityQueue")
25         priority, _, item = self._heap[0]
26         return item, priority
27
28     def __len__(self):
29         return len(self._heap)
30
31     def empty(self):
32         return len(self._heap) == 0
33
34
35 if __name__ == "__main__":
36     pq = PriorityQueue()
37     pq.push("low", 5)
38     pq.push("urgent", 1)
39     pq.push("medium", 3)
40     pq.push("also urgent (arrives later)", 1)
41
42     while not pq.empty():
43         item, pr = pq.pop()
44         print(pr, item)
45
46     # Max-priority example (bigger number = served first):
47     maxpq = PriorityQueue()
48     for item, pr in [("A", 10), ("B", 2), ("C", 10)]:
49         maxpq.push(item, -pr) # negate priority
50
51     print("max first:", maxpq.pop()) # returns (item, neg_priority)

```

Output:

```

1 urgent
1 also urgent (arrives later)
3 medium
5 low
max first: ('A', -10)

```

Task Description #8 – Deque

Task: Use AI to implement a double-ended queue using `collections.deque`.

Sample Input Code:

```

class DequeDS:
    pass

```

Prompt:

implement a double-ended queue using `collections.deque` with code

Code:

```

1  from __future__ import annotations
2
3  from collections import deque
4  from typing import Deque, Generic, Iterator, Optional, TypeVar
5
6  T = TypeVar("T")
7
8
9  class DequeDS(Generic[T]):
10     """
11     Double-ended queue (deque) implemented using collections.deque.
12     Supports O(1) append/pop operations on both ends.
13     """
14
15     def __init__(self, items: Optional[Iterator[T]] = None) -> None:
16         self._dq: Deque[T] = deque[T](items or [])
17
18     # --- Add operations ---
19     def add_front(self, item: T) -> None:
20         """Insert item at the front (left)."""
21         self._dq.appendleft(item)
22
23     def add_rear(self, item: T) -> None:
24         """Insert item at the rear (right)."""
25         self._dq.append(item)
26
27     # --- Remove operations ---
28     def remove_front(self) -> T:
29         """Remove and return the front (left) item."""
30         if self.is_empty():
31             raise IndexError("remove_front from empty deque")
32         return self._dq.popleft()
33
34     def remove_rear(self) -> T:
35         """Remove and return the rear (right) item."""
36         if self.is_empty():
37             raise IndexError("remove_rear from empty deque")
38         return self._dq.pop()
39
40     # --- Peek operations ---
41     def peek_front(self) -> T:
42         """Return the front (left) item without removing it."""
43         if self.is_empty():
44             raise IndexError("peek_front from empty deque")
45         return self._dq[0]
46
47     def peek_rear(self) -> T:
48         """Return the rear (right) item without removing it."""
49         if self.is_empty():
50             raise IndexError("peek_rear from empty deque")
51         return self._dq[-1]
52
53     # --- Utility ---
54     def is_empty(self) -> bool:
55         return len(self._dq) == 0
56
57     def size(self) -> int:
58         return len(self._dq)
59
60     def clear(self) -> None:
61         self._dq.clear()
62
63     def __len__(self) -> int:
64         return len(self._dq)
65
66     def __iter__(self) -> Iterator[T]:
67         return iter(self._dq)
68
69     def __repr__(self) -> str:
70         return f"DequeDS({list[T](self._dq)!r})"
71
72
73 if __name__ == "__main__":
74     d = DequeDS[int]()
75     d.add_front(10)    # [10]
76     d.add_rear(20)    # [10, 20]
77     d.add_front(5)    # [5, 10, 20]
78     print("Deque:", d)
79     print("Front:", d.peek_front())
80     print("Rear:", d.peek_rear())
81     print("Remove front:", d.remove_front()) # 5
82     print("Remove rear:", d.remove_rear())   # 20
83     print("Deque now:", d)

```

Output:

```
Deque: DequeDS([5, 10, 20])
Front: 5
Rear: 20
Remove front: 5
Remove rear: 20
Deque now: DequeDS([10])
```

Task Description #9 Real-Time Application Challenge – Choose the Right Data Structure

Prompt:

Solve this clearly and concisely.

Design a Campus Resource Management System code with:

Student Attendance Tracking

Event Registration System

Library Book Borrowing

Bus Scheduling System

Cafeteria Order Queue

Choose the best data structure for each feature from:

Stack, Queue, Priority Queue, Linked List, BST, Graph, Hash Table, Deque

Output as a table:

Feature | Data Structure | 2–3 sentence justification

Code:

[illegible]


```

104 book = self.find(book)
105 if not book:
106     return False
107 self.available_copies -= 1
108 book.available_copies += 1
109 return True
110
111 def catalog_in_order(self) -> list[Book]:
112     out: list[Book] = []
113     self._in_order(self._root, out)
114     return out
115
116 def _insert(self, node: Optional[BookNode], isbn: str, book: Book) -> BookNode:
117     if node is None:
118         return BookNode(isbn, book)
119     if isbn < node.isbn:
120         node.left = self._insert(node.left, isbn, book)
121     else:
122         node.right = self._insert(node.right, isbn, book)
123     return node
124
125 def _in_order(self, node: Optional[BookNode], out: list[Book]) -> None:
126     if node is None:
127         return
128     self._in_order(node.left, out)
129     out.append(node.book)
130     self._in_order(node.right, out)
131
132 # =====
133 # 4) Bus Scheduling System (Graph)
134 # =====
135
136 class BusNetwork:
137     """
138     Data structure: Graph (adjacency list)
139     - stops -> list of (stop_name, travel_minutes)
140     - shortest path uses Dijkstra (non-negative weights).
141     """
142
143     def __init__(self) -> None:
144         self._stops: list[tuple[str, int]] = []
145
146     def add_stop(self, stop: str) -> None:
147         self._stops.setdefault(stop, 0)
148
149     def add_route(self, s: str, t: str, minutes: int, bidirectional: bool = True) -> None:
150         if minutes < 0:
151             raise ValueError("minutes must be non-negative")
152         self._add_route(s, t, minutes)
153         self._add_route(t, s, minutes)
154         if bidirectional:
155             self._add_route(s, minutes)
156
157     def shortest_path(self, start: str, end: str) -> tuple[list, list[str]]:
158         if start not in self._stops or end not in self._stops:
159             raise ValueError("start/end stop not found")
160
161         dist: dict[str, int] = {start: 0}
162         prev: dict[str, Optional[str]] = {start: None}
163         pq: list[tuple[int, str]] = [(0, start)]
164
165         while pq:
166             d, u = heapq.heappop(pq)
167             if d != dist.get(u, 10**18):
168                 continue
169             if u == end:
170                 break
171             for v, w in self._stops[u]:
172                 nd = d + w
173                 if nd < dist.get(v, 10**18):
174                     dist[v] = nd
175                     prev[v] = u
176                     heapq.heappush(pq, (nd, v))
177
178         if end not in dist:
179             return (10**18, [])
180
181         # Reconstruct path
182         path: list[str] = []
183         cur: Optional[str] = end
184         while cur is not None:
185             path.append(cur)
186             cur = prev.get(cur)
187         path.reverse()
188         return (dist[end], path)
189
190 # =====
191 # 5) Cafeteria Order Queue (Priority Queue)
192 # =====
193
194 @dataclass(frozen=True)
195 class CafeteriaOrder:
196     order_id: int
197     student_id: str
198     item: str
199     priority: int # Higher number -> higher priority
200
201 class CafeteriaOrderSystem:
202     """
203     Data structure: Priority Queue (Heap)
204     - Serve highest priority first; tie-break by arrival order.
205     """
206
207     def __init__(self) -> None:
208         self._heap: list[tuple[int, int, str, CafeteriaOrder]] = []
209         self._counter = itertools.count()
210
211     def place_order(self, student_id: str, item: str, priority: int = 0) -> CafeteriaOrder:
212         order_id = next(self._counter)
213         order = CafeteriaOrder(order_id, student_id, item, priority)
214         # Insert in heap: lower priority or higher priority over first
215         heapq.heappush(self._heap, (-priority, order_id, order))
216         return order
217
218     def serve_next(self) -> Optional[CafeteriaOrder]:
219         if not self._heap:
220             return None
221         _, _, order = heapq.heappop(self._heap)
222         return order
223
224     def pending_count(self) -> int:
225         return len(self._heap)
226
227 # =====
228 # 6) Bus Network (Graph)
229 # =====
230
231 def main() -> None:
232     # Create
233     bus = BusNetwork()
234     bus.add_stop("A")
235     bus.add_stop("B")
236     bus.add_stop("C")
237     bus.add_stop("D")
238     bus.add_stop("E")
239     bus.add_stop("F")
240     bus.add_stop("G")
241     bus.add_stop("H")
242     bus.add_stop("I")
243     bus.add_stop("J")
244     bus.add_stop("K")
245     bus.add_stop("L")
246     bus.add_stop("M")
247     bus.add_stop("N")
248     bus.add_stop("O")
249     bus.add_stop("P")
250     bus.add_stop("Q")
251     bus.add_stop("R")
252     bus.add_stop("S")
253     bus.add_stop("T")
254     bus.add_stop("U")
255     bus.add_stop("V")
256     bus.add_stop("W")
257     bus.add_stop("X")
258     bus.add_stop("Y")
259     bus.add_stop("Z")
260     bus.add_stop("AA")
261     bus.add_stop("AB")
262     bus.add_stop("AC")
263     bus.add_stop("AD")
264     bus.add_stop("AE")
265     bus.add_stop("AF")
266     bus.add_stop("AG")
267     bus.add_stop("AH")
268     bus.add_stop("AI")
269     bus.add_stop("AJ")
270     bus.add_stop("AK")
271     bus.add_stop("AL")
272     bus.add_stop("AM")
273     bus.add_stop("AN")
274     bus.add_stop("AO")
275     bus.add_stop("AP")
276     bus.add_stop("AQ")
277     bus.add_stop("AR")
278     bus.add_stop("AS")
279     bus.add_stop("AT")
280     bus.add_stop("AU")
281     bus.add_stop("AV")
282     bus.add_stop("AW")
283     bus.add_stop("AX")
284     bus.add_stop("AY")
285     bus.add_stop("AZ")
286     bus.add_stop("BA")
287     bus.add_stop("BB")
288     bus.add_stop("BC")
289     bus.add_stop("BD")
290     bus.add_stop("BE")
291     bus.add_stop("BF")
292     bus.add_stop("BG")
293     bus.add_stop("BH")
294     bus.add_stop("BI")
295     bus.add_stop("BJ")
296     bus.add_stop("BK")
297     bus.add_stop("BL")
298     bus.add_stop("BM")
299     bus.add_stop("BN")
300     bus.add_stop("BO")
301     bus.add_stop("BP")
302     bus.add_stop("BQ")
303     bus.add_stop("BR")
304     bus.add_stop("BS")
305     bus.add_stop("BT")
306     bus.add_stop("BU")
307     bus.add_stop("BV")
308     bus.add_stop("BW")
309     bus.add_stop("BX")
310     bus.add_stop("BY")
311     bus.add_stop("BZ")
312     bus.add_stop("CA")
313     bus.add_stop("CB")
314     bus.add_stop("CC")
315     bus.add_stop("CD")
316     bus.add_stop("CE")
317     bus.add_stop("CF")
318     bus.add_stop("CG")
319     bus.add_stop("CH")
320     bus.add_stop("CI")
321     bus.add_stop("CJ")
322     bus.add_stop("CK")
323     bus.add_stop("CL")
324     bus.add_stop("CM")
325     bus.add_stop("CN")
326     bus.add_stop("CO")
327     bus.add_stop("CP")
328     bus.add_stop("CQ")
329     bus.add_stop("CR")
330     bus.add_stop("CS")
331     bus.add_stop("CT")
332     bus.add_stop("CU")
333     bus.add_stop("CV")
334     bus.add_stop("CW")
335     bus.add_stop("CX")
336     bus.add_stop("CY")
337     bus.add_stop("CZ")
338     bus.add_stop("DA")
339     bus.add_stop("DB")
340     bus.add_stop("DC")
341     bus.add_stop("DD")
342     bus.add_stop("DE")
343     bus.add_stop("DF")
344     bus.add_stop("DG")
345     bus.add_stop("DH")
346     bus.add_stop("DI")
347     bus.add_stop("DJ")
348     bus.add_stop("DK")
349     bus.add_stop("DL")
350     bus.add_stop("DM")
351     bus.add_stop("DN")
352     bus.add_stop("DO")
353     bus.add_stop("DP")
354     bus.add_stop("DQ")
355     bus.add_stop("DR")
356     bus.add_stop("DS")
357     bus.add_stop("DT")
358     bus.add_stop("DU")
359     bus.add_stop("DV")
360     bus.add_stop("DW")
361     bus.add_stop("DX")
362     bus.add_stop("DY")
363     bus.add_stop("DZ")
364     bus.add_stop("EA")
365     bus.add_stop("EB")
366     bus.add_stop("EC")
367     bus.add_stop("ED")
368     bus.add_stop("EE")
369     bus.add_stop("EF")
370     bus.add_stop("EG")
371     bus.add_stop("EH")
372     bus.add_stop("EI")
373     bus.add_stop("EJ")
374     bus.add_stop("EK")
375     bus.add_stop("EL")
376     bus.add_stop("EM")
377     bus.add_stop("EN")
378     bus.add_stop("EO")
379     bus.add_stop("EP")
380     bus.add_stop("EQ")
381     bus.add_stop("ER")
382     bus.add_stop("ES")
383     bus.add_stop("ET")
384     bus.add_stop("EU")
385     bus.add_stop("EV")
386     bus.add_stop("EW")
387     bus.add_stop("EX")
388     bus.add_stop("EY")
389     bus.add_stop("EZ")
390     bus.add_stop("FA")
391     bus.add_stop("FB")
392     bus.add_stop("FC")
393     bus.add_stop("FD")
394     bus.add_stop("FE")
395     bus.add_stop("FF")
396     bus.add_stop("FG")
397     bus.add_stop("FH")
398     bus.add_stop("FI")
399     bus.add_stop("FJ")
400     bus.add_stop("FK")
401     bus.add_stop("FL")
402     bus.add_stop("FM")
403     bus.add_stop("FN")
404     bus.add_stop("FO")
405     bus.add_stop("FP")
406     bus.add_stop("FQ")
407     bus.add_stop("FR")
408     bus.add_stop("FS")
409     bus.add_stop("FT")
410     bus.add_stop("FU")
411     bus.add_stop("FV")
412     bus.add_stop("FW")
413     bus.add_stop("FX")
414     bus.add_stop("FY")
415     bus.add_stop("FZ")
416     bus.add_stop("GA")
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418     bus.add_stop("GC")
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540     bus.add_stop("KU")
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566     bus.add_stop("LU")
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582     bus.add_stop("MK")
583     bus.add_stop("ML")
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592     bus.add_stop("MU")
593     bus.add_stop("MV")
594     bus.add_stop("MW")
595     bus.add_stop("MX")
596     bus.add_stop("MY")
597     bus.add_stop("MZ")
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599     bus.add_stop("NB")
600     bus.add_stop("NC")
601     bus.add_stop("ND")
602     bus.add_stop("NE")
603     bus.add_stop("NF")
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605     bus.add_stop("NH")
606     bus.add_stop("NI")
607     bus.add_stop("NJ")
608     bus.add_stop("NK")
609     bus.add_stop("NL")
610     bus.add_stop("NM")
611     bus.add_stop("NN")
612     bus.add_stop("NO")
613     bus.add_stop("NP")
614     bus.add_stop("NQ")
615     bus.add_stop("NR")
616     bus.add_stop("NS")
617     bus.add_stop("NT")
618     bus.add_stop("NU")
619     bus.add_stop("NV")
620     bus.add_stop("NW")
621     bus.add_stop("NX")
622     bus.add_stop("NY")
623     bus.add_stop("NZ")
624     bus.add_stop("OA")
625     bus.add_stop("OB")
626     bus.add_stop("OC")
627     bus.add_stop("OD")
628     bus.add_stop("OE")
629     bus.add_stop("OF")
630     bus.add_stop("OG")
631     bus.add_stop("OH")
632     bus.add_stop("OI")
633     bus.add_stop("OJ")
634     bus.add_stop("OK")
635     bus.add_stop("OL")
636     bus.add_stop("OM")
637     bus.add_stop("ON")
638     bus.add_stop("OO")
639     bus.add_stop("OP")
640     bus.add_stop("OQ")
641     bus.add_stop("OR")
642     bus.add_stop("OS")
643     bus.add_stop("OT")
644     bus.add_stop("OU")
645     bus.add_stop("OV")
646     bus.add_stop("OW")
647     bus.add_stop("OX")
648     bus.add_stop("OY")
649     bus.add_stop("OZ")
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652     bus.add_stop("PC")
653     bus.add_stop("PD")
654     bus.add_stop("PE")
655     bus.add_stop("PF")
656     bus.add_stop("PG")
657     bus.add_stop("PH")
658     bus.add_stop("PI")
659     bus.add_stop("PJ")
660     bus.add_stop("PK")
661     bus.add_stop("PL")
662     bus.add_stop("PM")
663     bus.add_stop("PN")
664     bus.add_stop("PO")
665     bus.add_stop("PP")
666     bus.add_stop("PQ")
667     bus.add_stop("PR")
668     bus.add_stop("PS")
669     bus.add_stop("PT")
670     bus.add_stop("PU")
671     bus.add_stop("PV")
672     bus.add_stop("PW")
673     bus.add_stop("PX")
674     bus.add_stop("PY")
675     bus.add_stop("PZ")
676     bus.add_stop("QA")
677     bus.add_stop("QB")
678     bus.add_stop("QC")
679     bus.add_stop("QD")
680     bus.add_stop("QE")
681     bus.add_stop("QF")
682     bus.add_stop("QG")
683     bus.add_stop("QH")
684     bus.add_stop("QI")
685     bus.add_stop("QJ")
686     bus.add_stop("QK")
687     bus.add_stop("QL")
688     bus.add_stop("QM")
689     bus.add_stop("QN")
690     bus.add_stop("QO")
691     bus.add_stop("QP")
692     bus.add_stop("QQ")
693     bus.add_stop("QR")
694     bus.add_stop("QS")
695     bus.add_stop("QT")
696     bus.add_stop("QU")
697     bus.add_stop("QV")
698     bus.add_stop("QW")
699     bus.add_stop("QX")
700     bus.add_stop("QY")
701     bus.add_stop("QZ")
702     bus.add_stop("RA")
703     bus.add_stop("RB")
704     bus.add_stop("RC")
705     bus.add_stop("RD")
706     bus.add_stop("RE")
707     bus.add_stop("RF")
708     bus.add_stop("RG")
709     bus.add_stop("RH")
710     bus.add_stop("RI")
711     bus.add_stop("RJ")
712     bus.add_stop("RK")
713     bus.add_stop("RL")
714     bus.add_stop("RM")
715     bus.add_stop("RN")
716     bus.add_stop("RO")
717     bus.add_stop("RP")
718     bus.add_stop("RQ")
719     bus.add_stop("RR")
720     bus.add_stop("RS")
721     bus.add_stop("RT")
722     bus.add_stop("RU")
723     bus.add_stop("RV")
724     bus.add_stop("RW")
725     bus.add_stop("RX")
726     bus.add_stop("RY")
727     bus.add_stop("RZ")
728     bus.add_stop("SA")
729     bus.add_stop("SB")
730     bus.add_stop("SC")
731     bus.add_stop("SD")
732     bus.add_stop("SE")
733     bus.add_stop("SF")
734     bus.add_stop("SG")
735     bus.add_stop("SH")
736     bus.add_stop("SI")
737     bus.add_stop("SJ")
738     bus.add_stop("SK")
739     bus.add_stop("SL")
740     bus.add_stop("SM")
741     bus.add_stop("SN")
742     bus.add_stop("SO")
743     bus.add_stop("SP")
744     bus.add_stop("SQ")
745     bus.add_stop("SR")
746     bus.add_stop("SS")
747     bus.add_stop("ST")
748     bus.add_stop("SU")
749     bus.add_stop("SV")
750     bus.add_stop("SW")
751     bus.add_stop("SX")
752     bus.add_stop("SY")
753     bus.add_stop("SZ")
754     bus.add_stop("TA")
755     bus.add_stop("TB")
756     bus.add_stop("TC")
757     bus.add_stop("TD")
758     bus.add_stop("TE")
759     bus.add_stop("TF")
760     bus.add_stop("TG")
761     bus.add_stop("TH")
762     bus.add_stop("TI")
763     bus.add_stop("TJ")
764     bus.add_stop("TK")
765     bus.add_stop("TL")
766     bus.add_stop("TM")
767     bus.add_stop("TN")
768     bus.add_stop("TO")
769     bus.add_stop("TP")
770     bus.add_stop("TQ")
771     bus.add_stop("TR")
772     bus.add_stop("TS")
773     bus.add_stop("TT")
774     bus.add_stop("TU")
775     bus.add_stop("TV")
776     bus.add_stop("TW")
777     bus.add_stop("TX")
778     bus.add_stop("TY")
779     bus.add_stop("TZ")
780     bus.add_stop("UA")
781     bus.add_stop("UB")
782     bus.add_stop("UC")
783     bus.add_stop("UD")
784     bus.add_stop("UE")
785     bus.add_stop("UF")
786     bus.add_stop("UG")
787     bus.add_stop("UH")
788     bus.add_stop("UI")
789     bus.add_stop("UJ")
790     bus.add_stop("UK")
791     bus.add_stop("UL")
792     bus.add_stop("UM")
793     bus.add_stop("UN")
794     bus.add_stop("UO")
795     bus.add_stop("UP")
796     bus.add_stop("UQ")
797     bus.add_stop("UR")
798     bus.add_stop("US")
799     bus.add_stop("UT")
800     bus.add_stop("UU")
801     bus.add_stop("UV")
802     bus.add_stop("UW")
803     bus.add_stop("UX")
804     bus.add_stop("UY")
805     bus.add_stop("UZ")
806     bus.add_stop("VA")
807     bus.add_stop("VB")
808     bus.add_stop("VC")
809     bus.add_stop("VD")
810     bus.add_stop("VE")
811     bus.add_stop("VF")
812     bus.add_stop("VG")
813     bus.add_stop("VH")
814     bus.add_stop("VI")
815     bus.add_stop("VJ")
816     bus.add_stop("VK")
817     bus.add_stop("VL")
818     bus.add_stop("VM")
819     bus.add_stop("VN")
820     bus.add_stop("VO")
821     bus.add_stop("VP")
822     bus.add_stop("VQ")
823     bus.add_stop("VR")
824     bus.add_stop("VS")
825     bus.add_stop("VT")
826     bus.add_stop("VU")
827     bus.add_stop("VV")
828     bus.add_stop("VW")
829     bus.add_stop("VX")
830     bus.add_stop("VY")
831     bus.add_stop("VZ")
832     bus.add_stop("WA")
833     bus.add_stop("WB")
834     bus.add_stop("WC")
835     bus.add_stop("WD")
836     bus.add_stop("WE")
837     bus.add_stop("WF")
838     bus.add_stop("WG")
839     bus.add_stop("WH")
840     bus.add_stop("WI")
841     bus.add_stop("WJ")
842     bus.add_stop("WK")
843     bus.add_stop("WL")
844     bus.add_stop("WM")
845     bus.add_stop("WN")
846     bus.add_stop("WO")
847     bus.add_stop("WP")
848     bus.add_stop("WQ")
849     bus.add_stop("WR")
850     bus.add_stop("WS")
851     bus.add_stop("WT")
852     bus.add_stop("WU")
853     bus.add_stop("WV")
854     bus.add_stop("WW")
855     bus.add_stop("WX")
856     bus.add_stop("WY")
857     bus.add_stop("WZ")
858     bus.add_stop("XA")
859     bus.add_stop("XB")
860     bus.add_stop("XC")
861     bus.add_stop("XD")
862     bus.add_stop("XE")
863     bus.add_stop("XF")
864     bus.add_stop("XG")
865     bus.add_stop("XH")
866     bus.add_stop("XI")
867     bus.add_stop("XJ")
868     bus.add_stop("XK")
869     bus.add_stop("XL")
870     bus.add_stop("XM")
871     bus.add_stop("XN")
872     bus.add_stop("XO")
873     bus.add_stop("XP")
874     bus.add_stop("XQ")
875     bus.add_stop("XR")
876     bus.add_stop("XS")
877     bus.add_stop("XT")
878     bus.add_stop("XU")
879     bus.add_stop("XV")
880     bus.add_stop("XW")
881     bus.add_stop("XX")
882     bus.add_stop("XY")
883     bus.add_stop("XZ")
884     bus.add_stop("YA")
885     bus.add_stop("YB")
886     bus.add_stop("YC")
887     bus.add_stop("YD")
888     bus.add_stop("YE")
889     bus.add_stop("YF")
890     bus.add_stop("YG")
891     bus.add_stop("YH")
892     bus.add_stop("YI")
893     bus.add_stop("YJ")
894     bus.add_stop("YK")
895     bus.add_stop("YL")
896     bus.add_stop("YM")
897     bus.add_stop("YN")
898     bus.add_stop("YO")
899     bus.add_stop("YP")
900     bus.add_stop("YQ")
901     bus.add_stop("YR")
902     bus.add_stop("YS")
903     bus.add_stop("YT")
904     bus.add_stop("YU")
905     bus.add_stop("YV")
906     bus.add_stop("YW")
907     bus.add_stop("YX")
908     bus.add_stop("YY")
909     bus.add_stop("YZ")
910     bus.add_stop("ZA")
911     bus.add_stop("ZB")
912     bus.add_stop("ZC")
913     bus.add_stop("ZD")
914     bus.add_stop("ZE")
915     bus.add_stop("ZF")
916     bus.add_stop("ZG")
917     bus.add_stop("ZH")
918     bus.add_stop("ZI")
919     bus.add_stop("ZJ")
920     bus.add_stop("ZK")
921     bus.add_stop("ZL")
922     bus.add_stop("ZM")
923     bus.add_stop("ZN")
924     bus.add_stop("ZO")
925     bus.add_stop("ZP")
926     bus.add_stop("ZQ")
927     bus.add_stop("ZR")
928     bus.add_stop("ZS")
929     bus.add_stop("ZT")
930     bus.add_stop("ZU")
931     bus.add_stop("ZV")
932     bus.add_stop("ZW")
933     bus.add_stop("ZX")
934     bus.add_stop("ZY")
935     bus.add_stop("ZZ")
936

```

Output:

```
Attendance S1 %: 50.0
Confirmed E100: ['S1', 'S2']
Waitlist E100: ['S3']
After cancel S2, confirmed: ['S1', 'S3']
Borrow Fluent Python: True
Borrow Fluent Python again: False
Catalog in order: [('9780134685991', 2), ('9781492051367', 0)]
Shortest bus path Hostel->Library: ['Hostel', 'Cafeteria', 'Library'] minutes: 9
Attendance S1 %: 50.0
Confirmed E100: ['S1', 'S2']
Waitlist E100: ['S3']
After cancel S2, confirmed: ['S1', 'S3']
Borrow Fluent Python: True
Borrow Fluent Python again: False
Catalog in order: [('9780134685991', 2), ('9781492051367', 0)]
Shortest bus path Hostel->Library: ['Hostel', 'Cafeteria', 'Library'] minutes: 9
Waitlist E100: ['S3']
After cancel S2, confirmed: ['S1', 'S3']
Borrow Fluent Python: True
Borrow Fluent Python again: False
Catalog in order: [('9780134685991', 2), ('9781492051367', 0)]
Shortest bus path Hostel->Library: ['Hostel', 'Cafeteria', 'Library'] minutes: 9
Borrow Fluent Python again: False
Catalog in order: [('9780134685991', 2), ('9781492051367', 0)]
Shortest bus path Hostel->Library: ['Hostel', 'Cafeteria', 'Library'] minutes: 9
Serve order: CafeteriaOrder(order_id=2, student_id='S2', item='Coffee', priority=2)
Shortest bus path Hostel->Library: ['Hostel', 'Cafeteria', 'Library'] minutes: 9
Serve order: CafeteriaOrder(order_id=2, student_id='S2', item='Coffee', priority=2)
Serve order: CafeteriaOrder(order_id=2, student_id='S2', item='Coffee', priority=2)
Serve order: CafeteriaOrder(order_id=3, student_id='S3', item='Burger', priority=1)
Serve order: CafeteriaOrder(order_id=3, student_id='S3', item='Burger', priority=1)
```

Task Description #10: Smart E-Commerce Platform – Data Structure

Prompt:

Solve this clearly and concisely.

Design a Smart E-Commerce Platform with:

Shopping Cart Management – Add/remove products dynamically

Order Processing System – Process orders in placement order

Top-Selling Products Tracker – Rank products by sales count

Product Search Engine – Fast lookup using product ID

Delivery Route Planning – Connect warehouses and delivery locations

Choose the most appropriate data structure for each feature from:

Stack, Queue, Priority Queue, Linked List, BST, Graph, Hash Table, Deque

Output as a table:

Feature | Data Structure | 2–3 sentence justification

Code:

```

1 from collections import deque
2 import heapq
3 from typing import Dict, List, Tuple, Optional
4
5
6 # -----
7 # Product model
8 # -----
9
10 class Product:
11     def __init__(self, product_id: int, name: str, price: float):
12         self.id = product_id
13         self.name = name
14         self.price = price
15
16     def __repr__(self):
17         return f"Product(id={self.id}, name='{self.name}', price={self.price})"
18
19 # -----
20 # Product Search Engine (Hash Table)
21 # -----
22
23 class ProductSearchEngine:
24     def __init__(self):
25         # Hash Table: product_id -> Product
26         self.products: Dict[int, Product] = {}
27
28     def add_product(self, product: Product):
29         self.products[product.id] = product
30
31     def get_product(self, product_id: int) -> Optional[Product]:
32         return self.products.get(product_id)
33
34     def remove_product(self, product_id: int):
35         self.products.pop(product_id, None)
36
37 # -----
38 # Shopping Cart (Linked List)
39 # -----
40
41 class CartNode:
42     def __init__(self, product: Product, quantity: int):
43         self.product = product
44         self.quantity = quantity
45         self.next: Optional[CartNode] = None
46
47 class ShoppingCart:
48     def __init__(self):
49         self.head: Optional[CartNode] = None
50
51     def add_product(self, product: Product, quantity: int = 1):
52         """
53         If product already exists in the list, increase quantity.
54         Otherwise, add new node at the front (O(1) insertion).
55         """
56         node = CartNode(product, quantity)
57         while node:
58             if node.product.id == product.id:
59                 node.quantity += quantity
60                 return
61             node = node.next
62         new_node = CartNode(product, quantity)
63         new_node.next = self.head
64         self.head = new_node
65
66     def remove_product(self, product_id: int, quantity: int = None):
67         """
68         Remove some or all quantity of a product.
69         If quantity is None or reaches 0, remove the node.
70         """
71         prev = None
72         node = self.head
73         while node:
74             if node.product.id == product_id:
75                 if quantity is None or node.quantity <= quantity:
76                     # delete the node
77                     if prev:
78                         prev.next = node.next
79                     else:
80                         self.head = node.next
81                 else:
82                     node.quantity -= quantity
83                     return
84                 prev = node
85                 node = node.next
86
87     def list_items(self) -> List[Tuple[Product, int]]:
88         result = []
89         node = self.head
90         while node:
91             result.append((node.product, node.quantity))
92             node = node.next
93         return result
94
95     def total_price(self) -> float:
96         return sum(node.product.price * node.quantity
97                     for node in self._iter_nodes())
98
99     def _iter_nodes(self):
100         node = self.head
101         while node:
102             yield node
103             node = node.next
104
105 # -----
106 # Order Processing System (Queue)
107 # -----
108
109 class Order:
110     __next_id = 1
111
112     def __init__(self, cart_snapshot: List[Tuple[Product, int]]):
113         self.id = Order.__next_id
114         Order.__next_id += 1
115         self.items = cart_snapshot # list of (Product, quantity)
116
117     def __repr__(self):
118         return f"Order(id={self.id}, items=[{(p.id, q) for p, q in self.items}])"
119
120 class OrderProcessingSystem:
121     def __init__(self):
122         # Queue of orders (FIFO)
123         self.queue: deque[Order] = deque[Order]()
124
125     def place_order(self, cart: ShoppingCart) -> Order:
126         order = Order(cart.list_items())
127         self.queue.append(order)
128         return order
129
130     def process_next_order(self) -> Optional[Order]:
131         if not self.queue:
132             return None
133         return self.queue.popleft()
134
135     def pending_orders(self) -> int:
136         return len(self.queue)
137
138 # -----
139 # Top-Selling Products Tracker (Priority Queue / Max-Heap)
140 # -----
141
142 class TopSellingProductsTracker:
143     def __init__(self):
144         # product_id -> sales_count
145         self.sales: Dict[int, int] = {}
146         # priority queue entries: (-sales_count, product_id)
147         self.heap: List[Tuple[int, int]] = []
148
149     def record_sale(self, product_id: int, quantity: int = 1):
150         self.sales[product_id] = self.sales.get(product_id, 0) + quantity
151         # Push new priority entry; lazy update (we'll verify against self.sales on pop)
152         heapq.heappush(self.heap, (-self.sales[product_id], product_id))
153
154     def top_k(self, k: int) -> List[Tuple[int, int]]:
155         """
156         Returns list of (product_id, sales_count) for top k products.
157         Uses lazy removal from the heap to keep it consistent.
158         """
159 
```

```

162         result = []
163         seen = set()
164
165         while self.heap and len(result) < k:
166             neg_sales, pid = heapq.heappop(self.heap)
167             current_sales = self.sales.get(pid, 0)
168
169             if current_sales == -neg_sales and pid not in seen:
170                 result.append((pid, current_sales))
171                 seen.add(pid)
172
173             # push back the elements we popped that are still valid
174             for pid in seen:
175                 heapq.heappush(self.heap, (-self.sales[pid], pid))
176
177         return result
178
179
180 # =====
181 # Delivery Route Planning (Graph + Dijkstra)
182 # =====
183 class DeliveryRoutePlanner:
184     def __init__(self):
185         # Graph as adjacency list: node -> (list of (neighbor, distance))
186         self.graph = Dict[str, List[Tuple[str, float]]] = {}
187
188     def add_location(self, name: str):
189         if name not in self.graph:
190             self.graph[name] = []
191
192     def add_route(self, from_loc: str, to_loc: str, distance: float, bidirectional: bool = True):
193         self.add_location(from_loc)
194         self.add_location(to_loc)
195         self.graph[from_loc].append((to_loc, distance))
196         if bidirectional:
197             self.graph[to_loc].append((from_loc, distance))
198
199     def shortest_path(self, start: str, end: str) -> Tuple[float, List[str]]:
200         # Dijkstra's algorithm: returns (distance, path).
201         # Distance is float('inf') if no path exists.
202         if start not in self.graph or end not in self.graph:
203             return float('inf'), []
204
205         # min-heap: (distance, node, path)
206         heap = [(0.0, start, [start])]
207         visited = set()
208
209         while heap:
210             dist, node, path = heapq.heappop(heap)
211             if node in visited:
212                 continue
213             visited.add(node)
214
215             if node == end:
216                 return dist, path
217
218             for neighbour, weight in self.graph[node]:
219                 if neighbour not in visited:
220                     new_dist = dist + weight
221                     new_path = path + [neighbour]
222                     heapq.heappush(heap, (new_dist, neighbour, new_path))
223
224         return float('inf'), []
225
226 # =====
227 # Example usage
228 # =====
229 if __name__ == "__main__":
230     # Product search engine
231     search_engine = ProductSearchEngine()
232     p1 = Product(1, "Laptop", 1000.0)
233     p2 = Product(2, "Phone", 500.0)
234     p3 = Product(3, "Headphones", 100.0)
235     for p in (p1, p2, p3):
236         search_engine.add_product(p)
237
238     # Shopping cart
239     cart = ShoppingCart()
240     cart.add_product(search_engine.get_product(1), 1)
241     cart.add_product(search_engine.get_product(2), 2)
242     cart.add_product(search_engine.get_product(3), 3)
243     cart.remove_product(3, 1) # remove 1 headphone
244
245     print("Cart items:", cart.list_items())
246     print("Total price:", cart.total_price())
247
248     # Order processing
249     ops = OrderProcessingSystem()
250     order1 = ops.place_order(cart)
251     print("Placed order:", order1)
252     print("Pending orders:", ops.pending_orders())
253     processed = ops.process_next_order()
254     print("Processed order:", processed)
255     print("Pending orders:", ops.pending_orders())
256
257     # Top-selling products
258     tracker = TopSellingProductsTracker()
259     tracker.record_sale(1, 10) # Laptop sold 10
260     tracker.record_sale(2, 5) # Phone sold 5
261     tracker.record_sale(3, 7) # Headphones sold 7
262     print("Top 2 products (id, sales):", tracker.top_k(2))
263
264     # Delivery route planner
265     planner = DeliveryRoutePlanner()
266     planner.add_route("WarehouseA", "City1", 10.0)
267     planner.add_route("WarehouseA", "City2", 20.0)
268     planner.add_route("City1", "City2", 5.0)
269     planner.add_route("City2", "City3", 7.0)
270
271     dist, path = planner.shortest_path("WarehouseA", "City3")
272     print("Shortest route WarehouseA -> City3: ", path, "distance:", dist)
273

```

Output:

```

Cart items: [(Product(id=3, name='Headphones', price=100.0), 2), (Product(id=2, name='Phone', price=500.0), 2), (Product(id=1, name='Laptop', price=1000.0), 1)]
Total price: 2200.0
Placed order: Order(id=1, items=[(3, 2), (2, 2), (1, 1)])
Pending orders: 1
Processed order: Order(id=1, items=[(3, 2), (2, 2), (1, 1)])
Pending orders: 0
Top 2 products (id, sales): [(1, 10), (3, 7)]
Shortest route WarehouseA -> City3: ['WarehouseA', 'City1', 'City2', 'City3'] distance: 22.0
PS C:\2403A51103\3-2\AI_A_C\cursor AI>

```

```

Total price: 2200.0
Placed order: Order(id=1, items=[(3, 2), (2, 2), (1, 1)])
Pending orders: 1
Processed order: Order(id=1, items=[(3, 2), (2, 2), (1, 1)])
Pending orders: 0
Top 2 products (id, sales): [(1, 10), (3, 7)]
Shortest route WarehouseA -> City3: ['WarehouseA', 'City1', 'City2', 'City3'] distance: 22.0
Pending orders: 0
Top 2 products (id, sales): [(1, 10), (3, 7)]
Shortest route WarehouseA -> City3: ['WarehouseA', 'City1', 'City2', 'City3'] distance: 22.0
Shortest route WarehouseA -> City3: ['WarehouseA', 'City1', 'City2', 'City3'] distance: 22.0

```